# Performance Estimation Problems (PEPs)

## Systematic, principled, and computer-aided approaches to the analysis and design of first-order optimization algorithms

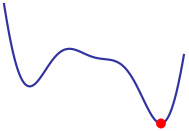
Aymeric Dieuleveut, Adrien Taylor

SMAI-MODE tutorial – March 2026

# Context: numerical (continuous) optimization

Minimize $f : \mathbb{R}^d \to \mathbb{R}$ (e.g., with $f$ continuous)

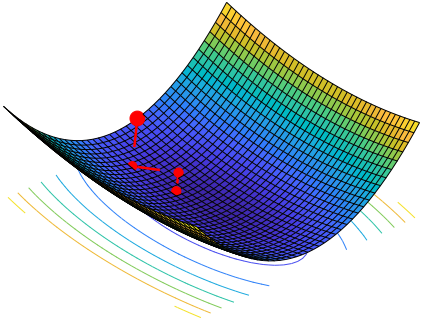
$$f(x_\star) \triangleq \min_{x \in \mathbb{R}^d} f(x).$$



**Ubiquitous in applied mathematics and computer science.**

Numerous applications for modeling (physics, economics), estimation (statistics, machine learning), decisions (control, operations research).

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.



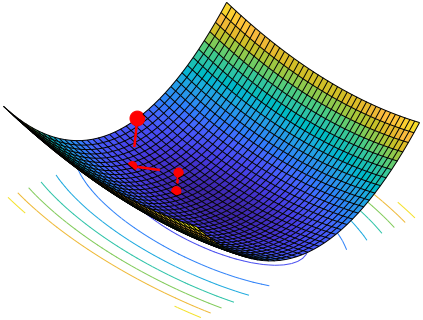
**Gradient descent** (stepsize $\alpha$)

**for** $k = 0, 1, \ldots$ **do**

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

**end for**

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.



**Gradient descent** (stepsize $\alpha$)

**for** $k = 0, 1, \ldots$ **do**

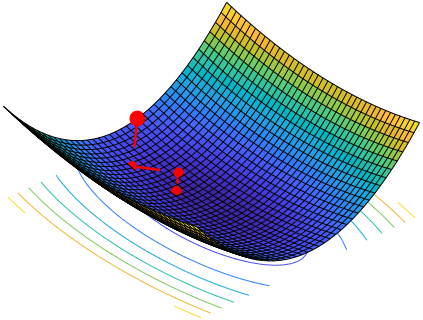
$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

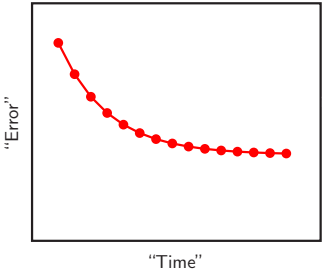
**end for**

What to expect from the output of the algorithm?

For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.

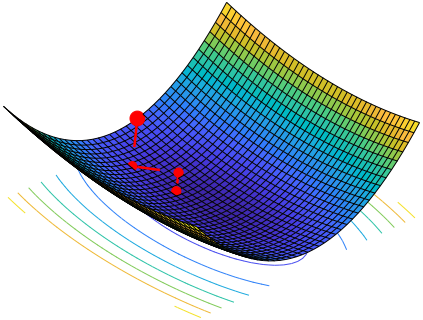


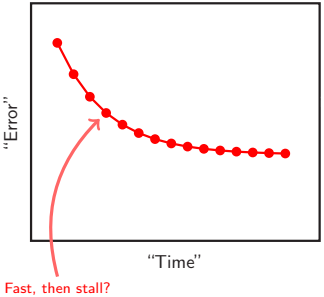



What to expect from the output of the algorithm?

For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.

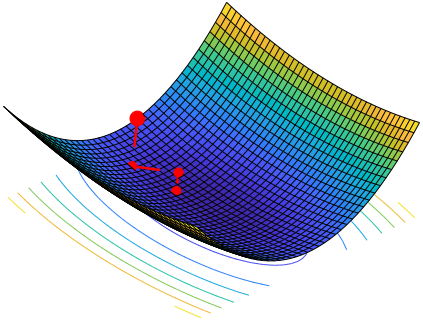


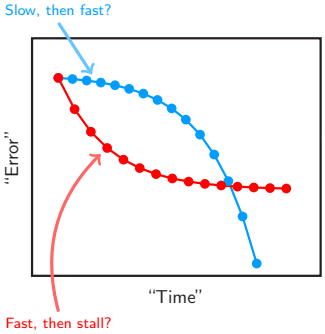



What to expect from the output of the algorithm?

For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.

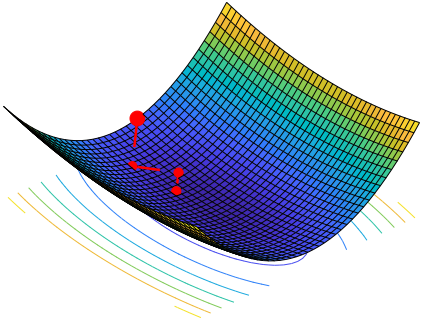


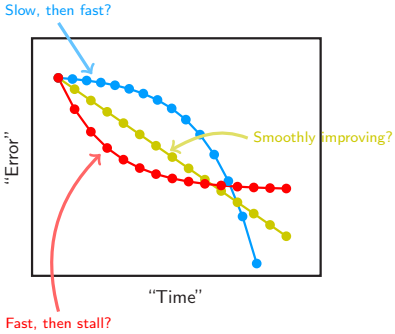



What to expect from the output of the algorithm?

For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.

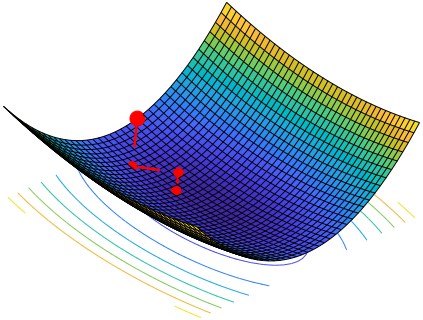


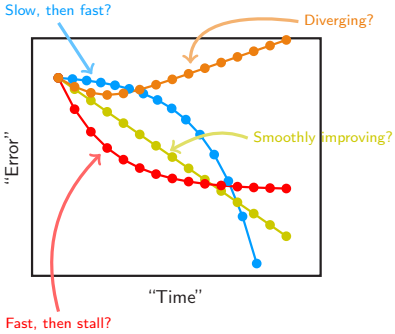



What to expect from the output of the algorithm?

For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

Usually solved via **iterative algorithm** generating sequence $x_0, x_1, \ldots, x_N$.






What to expect from the output of the algorithm?

For instance: **bounds** on certain notions of "error": $f(x_k) - f(x_\star)$, $\|x_k - x_\star\|$, $\|\nabla f(x_k)\|$, etc.

# How to show that an algorithm works?

# How to show that an algorithm works?

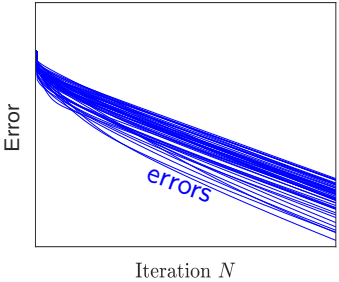
- Assumptions (no free lunch).

# How to show that an algorithm works?

- Assumptions (no free lunch).
- Here: worst-case perspective.

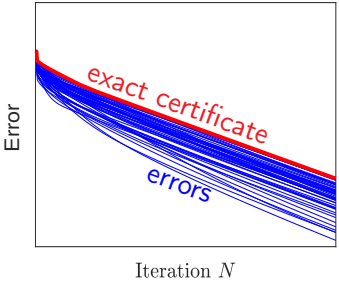
# How to show that an algorithm works?

- Assumptions (no free lunch).
- Here: worst-case perspective.



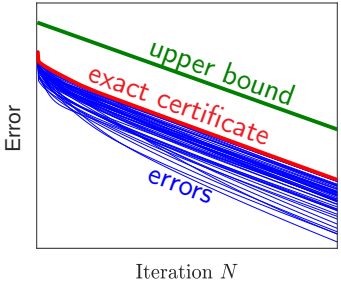
# How to show that an algorithm works?

- Assumptions (no free lunch).
- Here: worst-case perspective.



# How to show that an algorithm works?

- Assumptions (no free lunch).
- Here: worst-case perspective.



# Organization and learning outcomes (1/3)

**Observations:**

- ◇ Complexity analyses / convergence proofs for first-order optimization methods often follow very similar (obscure?) patterns.
- ◇ ... these proofs are rarely intuitive—rely on combining (many) inequalities.
- ◇ Many variations around same themes.

*Performance estimation framework*:

- ◇ helps understand proof structures of worst-case analyses,
- ◇ provides principled way to explore complexity bounds/derive such analyses,
- ◇ mathematical proofs via numerical experiments and symbolic computations.

# Organization and learning outcomes (2/3)

**This mini-course:**

- ◇ Interactive introduction to *performance estimation problems* (PEPs).
- ◇ Learn how to use numerical PEPs to assess performance of simple first-order algorithms.
- ◇ Learn how to exploit PEP outputs to construct Lyapunov-based convergence proofs.
- ◇ Symbolic computations for search & verifying obtained guarantees.
- ◇ Learn the basics of PEPs for numerical design of optimization algorithms.

# Organization and learning outcomes (3/3)

0. Organization (now)
1. Base performance estimation problems
   - Simple example
   - first-order oracles, duality, interpolation, semidefinite programming.
2. Structured convergence & non-convergence certificates
   - Using PEPs to search for structured proofs.
   - Using PEPs to search for cycles (no convergence). (convergence).
3. Numerical design of optimized algorithms
   - Using PEPs to design algorithms.

For all parts: examples, guided hands-on notebooks, and Q&A.

# Ressources

`https://github.com/PerformanceEstimation/Tutorial-SMAI-MODE`

Tools:

- Jupyter notebooks,
- Optimization tools/packages:
  - convex optimization interfacing/modeling tool: CVXPY.
  - Semidefinite (SDP) solvers: get academic MOSEK if possible (here).
- Symbolic computations:
  - SymPy [**OK for this mini-tutorial**], or Wolfram Engine (here+jupyter).
  - Mathematica is better (but expensive) — free (limited) version via cloud: here.

Do interrupt with questions! Pace & program probably optimistic and to be adapted.